

Fran Menéndez Moya

+34 678 77 91 50 fmenendezmoya@gmail.com linkedin.com/in/fmenemo github.com/fmenemo fmenemo.github.io
Zaragoza, Spain | EU work authorization | Remote

Summary

Full-stack engineer with 10+ years shipping whole products end to end in TypeScript, from the database through the API to the interface and the pipeline that releases it. I work at the AI layer and across the full stack: semantic search in production, MCP tooling I made the team-wide standard, and an agentic engineering workflow I built and run at The Knot for production delivery. I lead architecture domain-wide, rebuild the systems holding a product back, set standards the team adopts, and mentor engineers into ownership.

Technical Skills

AI Engineering:	Agentic Development Workflow (role-scoped agents, dependency-derived scheduling, model & effort routing, schema-typed review gates, bounded retries), MCP (tooling & design-to-code), Semantic Search (OpenSearch k-NN, vector embeddings, BM25)
Frontend:	Next.js, React, Vue.js, Angular, Astro
Backend:	Node.js, NestJS, Microservices, Real-time Systems, REST, GraphQL
Data & Search:	PostgreSQL, MongoDB, CouchDB, OpenSearch, Redis
Application Security:	SQL injection prevention, access control & IDOR, PII exposure, security regression testing in CI
Infrastructure:	AWS (Cognito, Lambda, CloudFront, S3), Kubernetes, Docker, CDN, Message Queues
CI/CD & Tools:	Jenkins, GitHub Actions, CircleCI, Fastlane, Git, Automated Pipelines
CMS & Commerce:	PayloadCMS, Contentful (incl. custom editor apps), Headless Architecture
Mobile & Cross-Platform:	React Native, Ionic, Cordova, Electron, PWAs

Professional Experience

The Knot Worldwide

Oct 2023 - Jul 2026

Remote

Principal Software Engineer

Apr 2025 - Jul 2026

- Took Shop from proposal to production: the e-commerce platform I proposed as Lead, on a \$2M+ annual revenue projection, delivered to a live MVP with the team, and took to production as Principal - sole contributor on it at times. Where the build order was contested I argued it from what the product's metrics showed rather than from the surface being asked for.
 - After the mobile-only MVP the plan was the desktop build next; I argued from Mixpanel and GA, an onboarding drop-off on a mostly mobile audience, that retention on mobile came first, and that order was adopted.
 - Migrated PayloadCMS from v2 to v3 for 100,000+ products with zero downtime; query response times dropped from 850ms to 34ms through database optimization and a Redis query cache.
 - Rebuilt the Shop e-commerce platform from scratch in one month solo (versus a three-month, multi-developer estimate). The platform handles millions of products from Amazon's Rainforest API and editorial content via Contentful, built with Next.js and React with shared component libraries, now serving hundreds of thousands of users in production.
 - Built a semantic product matching system using OpenSearch with k-NN vector similarity and BM25 text relevance, banded by confidence: candidates below a 0.75 floor were discarded, the strongest matches served automatically - about 90% of throughput, with no human review - and the band in between routed to a custom review dashboard I built inside PayloadCMS. p95 query latency stayed under 50ms across 100,000+ products.
 - Building on that migration, reworked caching for 2M+ weekly users: added a cache layer and set per-endpoint revalidation and staleness policies after investigating each endpoint's tolerance, on the existing CDN with no added infrastructure. The reduced origin load let the pods be right-sized down, cutting the resource allocated and billed per pod with no availability regression.
 - Extended the PayloadCMS admin with custom pages: a purge console giving editors control over that Redis cache without an engineer, and the manual review panel for the product matcher.
 - Unified sign-on across five products - Bump articles, baby names, registry, shop and the native apps - so that one account replaced five separate logins.
 - Built session continuity across the Bump platform and Shop boundary: an OAuth token handoff signs users into Shop with their existing Bump account on navigation, provisioning an account just-in-time when none exists.
 - Hardened the public e-commerce API: closed SQL injection, access-control and PII-exposure holes, including a write-side IDOR in a shared authorisation primitive covering five collections, and left it guarded by regression tests that run in CI.
- Introduced Model Context Protocol (MCP) tooling that generates production components directly from Figma at token-exact fidelity, removing the manual design-to-code translation step; now the team-wide standard. On my own implementation work it took design-with-components from about a week to under a day, roughly 85%.
- Built and ran an agentic AI development workflow for my own production delivery: four stages, eight role-scoped agents, model routing by task, a thin-orchestrator context budget, and review roles barred from writing the code they audit. Used it to deliver a security hardening programme, where the independent step caught defects the implementing pass had missed; drove its practices into the team's process.

- Steadied the engineering team through a company restructuring, and argued build-capacity against target-attainment in the half-year OKR capacity review; adjudicated review disputes on evidence rather than seniority.
- Built a custom Contentful editor app for the editorial platform (React, Contentful App SDK), including its S3 + CloudFront hosting infrastructure; propagated the new content model end-to-end across three services with zero-downtime migrations.
- Traced a sitewide collapse in The Bump's organic traffic to the site taking its own address from whichever host the request arrived on. An ingress migration had just changed that host, so the site was telling search engines its internal origin was canonical. Pinned that identity to the brand domain where it is derived, so no later infrastructure change can move it.

Lead Software Engineer

Oct 2023 - Mar 2025

- Took over The Bump's web platform from the outgoing team: rebuilt their build and release tooling, environments and runbooks in-house, and sat on the hiring panel for four engineering roles, defining the technical screen, to staff the team that would own it.
- Built an A/B testing system for The Bump, a platform where edge caching had made experimentation impossible: Akamai assigns a variant cookie at the edge, the app reads it to render the matching tagged build, and assignment stays cached and sticky across reloads until the cookie expires. The platform had run no experiments in over 3 years before it; 10+ have run on it since.
- The winning variant in an ad-layout test on that system lifted served ad impressions 23% against control on an ad-monetised property; it shipped because engagement depth held at baseline in Mixpanel and GA4, with the new ad combination sitting outside the width-capped article column.
- Managed an 8-person cross-functional team; mentored 5 developers, resulting in 2 promotions, with 100% retention across the period.
- Cut deployment time from over 2 hours to 1 minute, taking releases from batched to daily.

MOBIKO GmbH

Aug 2020 - Sep 2023

Munich, Germany (remote)

Team Lead & Architecture

Mar 2023 - Sep 2023

- Set the technical strategy and built the microservices architecture behind the company's enterprise expansion - three global enterprise partnerships, each scoped directly with the customer and integrated per partner - processing 500k+ daily mobility events: a NestJS API gateway fronting serverless functions communicating over NATS message queues, containerized into Kubernetes with Keda-driven autoscaling and GitHub Actions deployment workflows.
- Restructured the review and release process and mentored engineers into ownership; the team grew and delivery cycles shortened over the period.

Senior Full-Stack Developer

Aug 2020 - Mar 2023

- Rebuilt the API surface on NestJS with Swagger-annotated contracts and TypeORM against PostgreSQL, cutting API response times 72% through caching and database query optimization; page loads dropped 70% in Lighthouse.
- Scaled the user base from 3,000 to 10,000+ users (233% growth) through performance optimization and strategic feature development.
- Migrated the entire product onto a new Vue platform and built the PWA intended to replace the existing multiplatform applications, on a Vuex/Flux state architecture in a class-structured TypeScript codebase.
- Automated the deployment pipeline from multi-day release cycles to continuous deployment, on Kubernetes with CircleCI running automated deploys and compatibility checks.

Hiberus Tecnología

Jul 2017 - Jul 2020

Zaragoza, Spain

Senior Full-Stack Developer

Jan 2020 - Jul 2020

- Rebuilt the checkout path for a real-time ticketing platform under La Liga match-day load, integrating the official API for 100k+ concurrent users.
- Built Afición360, a single Angular monorepo serving the front ends of multiple football clubs plus an internal admin dashboard, on a shared Flux state architecture with NestJS REST microservices split via Lerna; integrated the La Liga and AVET APIs, using websockets deliberately to throttle traffic so the external services were not overloaded.
- Built and configured Jenkins from scratch, with pipelines building and deploying every monorepo application and backend into Docker on AWS.

Full-Stack Developer

Nov 2017 - Jan 2020

- Built a platform processing 1M+ daily entries with real-time sync across 5 geographic locations, on CouchDB as an offline-capable distributed database live-syncing data spread across multiple systems, at 99.95% uptime; still in production.
- Built cross-platform Ionic applications that ran fully offline or embedded existing web apps, using Cordova for native device capabilities (bluetooth, GPS, push notifications), and compiled Angular applications into Electron to run as kiosks; automated mobile build, test and marketplace deployment with Fastlane.

Junior Developer, E-commerce

Jul 2017 - Nov 2017

- Built a NodeJS/Express service from scratch aggregating external APIs, still in production, and an AngularJS application for a multinational consuming Google ecosystem APIs; backend work on an in-house Spring/Hibernate framework over MySQL and SQL Server.

Earlier: Freelance Web Developer

2016 - Jul 2017

- Delivered a full redesign across frontend, backend, database and self-hosted deployment for a university graduation photography studio: a public interface where students booked appointments to have their graduation photographs taken, plus an administration panel managing those appointments and private events.

Independent Work

Instagram Checker

Live and built end to end with the same agentic workflow: it turns an Instagram data export the user supplies into a follower and following dashboard, reading only what the user provides rather than scraping.

Multi-agent delivery harness

Built and run solo. Scheduling comes from the tracker's dependency edges, unblocked work in parallel. An independent review answers in a schema the harness parses, and the parsed verdict gates the merge; a finding returns to the writing session as a bounded retry. Every step lands on a typed transition stream, and each role runs at the model and effort its failure cost argues for.

Education & Languages

Computer Engineering (Software Engineering Specialization) Universidad de Zaragoza

2012-2017

Languages

Spanish (Native) | English (Advanced, Cambridge C1, certificate 14CES0055005)

Recognitions & Awards

Google Hash Code Zaragoza Hub Winner	Feb 2019
NASA Space Apps Zaragoza Local Hub Winner	Oct 2018
ImagineCode Blockchain Challenge Winner	Oct 2018
uCode by Adidas, Best Fan Experience Solution	Mar 2018
100 Ideas Zaragoza, Best Use of the Citizen Card	Sep 2017
100 Ideas Zaragoza, Most Innovative Technology	Sep 2017
NASA Space Apps Global Finalist & Galactic Impact Award	May 2017